project-36

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1 Review Methodology and the Secondary Evidence Base

Ten of the sixty-six studies in this corpus (15.2%) are reviews, surveys, or conceptual syntheses. They constitute evidence about the field's methodology rather than instances of it, and they supply the vocabulary in which the remaining fifty-six studies are described. Seven systematic or scoping reviews of large language models (LLMs) in mental health appeared between 2024 and 2026, so a further synthesis must show what those seven leave unaddressed. The ten papers fall into three groups: scoping reviews that map breadth, systematic reviews conducted under PRISMA that also attempt appraisal, and conceptual papers that supply taxonomies and governance frameworks without primary data.

1.1 Scoping reviews

Four studies adopt a scoping design aimed at coverage rather than effect estimation.

Jin et al.[25] searched seven databases, including the Chinese-language Weipu, CNKI and Wanfang, for work published between 1 January 2019 and 31 August 2024, including 95 of 4,859 retrieved records. Two features distinguish it: prospective registration on the Open Science Framework, adopted by only two of the seven empirical reviews, and systematic non-English coverage, which matters given the volume of primary work from Chinese institutions. Screening proceeded in three rounds with disagreements escalated to third reviewers, and extracted data were synthesised descriptively in R 4.4.2 without pooling.

Hua et al.[21] searched six sources, three of them preprint servers, covering 1 October 2019 to 2 December 2023, and included 34 of 313 records. Admitting preprints reduces the recency lag affecting every other review here at the cost of unrefereed work. GPT-4 served as a parallel screening reviewer alongside a human, with agreement quantified by Cohen's κ (≈ 0.90). Among the included reviews, this is the sole instance in which LLM-assisted screening is itself quantified rather than assumed.

Yang et al.[64] retrieved 10,743 records across eleven databases, assessed 58 at full text and charted 29, following both PRISMA-ScR and the Joanna Briggs Institute framework. Two independent reviewers screened and charted, with a third resolving discrepancies. Where the other reviews ask what LLMs can do, this one asks where their application boundaries lie, charting technical, ethical and practical limits per study.

Hua et al.[22] searched four databases for peer-reviewed work published between 1 January 2020 and 19 July 2024, including 16 of 726 unique articles under PRISMA 2020. Its narrower scope, generative tasks only under a parameter threshold, yields the smallest included set among the scoping reviews. Mapping primary studies onto a three-level clinical evaluation pyramid converts a descriptive exercise into a diagnostic one and shows that the literature clusters at the lowest tier.

1.2 PRISMA systematic reviews

Three studies adopt a systematic design, and the differences between them are instructive precisely because the label is shared.

Wang et al.[58] screened 1,046 references across five databases with paired independent reviewers in Rayyan and included eight studies, an inclusion rate of 0.76%. It applies the Mixed Methods Appraisal Tool, one of two reviews here to use a formal appraisal instrument. Primary research is classified by the clinical competencies a human therapist would require, which organises model evaluation around clinical role rather than technical task. With $n = 8$, and with included studies weighted toward zero-shot prompt tests, its conclusions rest on a thin base.

Kolding et al.[30] screened 724 unique records after removing 432 duplicates from 1,156, including 40 studies under a protocol preregistered on the Open Science Framework, with dual-reviewer screening at both stages supported by Covidence. No formal quality appraisal or quantitative synthesis was performed, which the authors attribute to the immaturity of the field rather than to resource constraints. Their observation that models and prompts are frequently underspecified to the point where replication is impossible is consistent with the extraction reported in Section 7.

Guo et al.[17] collated 40 peer-reviewed articles from five databases covering 1 January 2017 to 30 April 2024, the widest window among the seven. It applies the Cochrane Risk of Bias 2 tool and is one of two reviews to report inter-reviewer agreement (Cohen's $\kappa = 0.84$). Current risks of clinical use are judged to outweigh the benefits, with the absence of a benchmarked ethical framework identified as a gap alongside hallucination and output inconsistency. A stated limitation is that 98% of reviewed papers cover BERT- and GPT-family models alone.

Table 1.1: Review protocol parameters across the seven empirical reviews

Study	Design	Databases	Screened → Included	Search window ends	Dual screening	Quality appraisal	Agreement reported	Pooling
Jin et al.[25]	Scoping	7 (incl. 3 Chinese)	4,859 → 95	Aug 2024	Yes (3 rounds)	No	No	None
Hua et al.[21]	Scoping	6 (incl. preprints)	313 → 34	Dec 2023	Human + GPT-4	No	$\kappa \approx 0.90$	None
Yang et al.[64]	Scoping (PRISMA-ScR + JBI)	11	10,743 → 29	Not reported	Yes (2 + adjudicator)	No	No	None
Hua et al.[22]	Scoping (PRISMA 2020)	4	726 → 16	Jul 2024	Yes	No	No	None
Wang et al.[58]	Systematic (PRISMA)	5	1,046 → 8	Jun 2024	Yes (paired)	MMAT	No	None
Kolding et al.[30]	Systematic (PRISMA, preregistered)	3	724 → 40	Feb 2024	Yes (both stages)	Declined, stated	No	None
Guo et al.[17]	Systematic (PRISMA)	5	Not reported → 40	Apr 2024	Yes (2 + adjudicator)	RoB 2	$\kappa = 0.84$	None

1.3 Conceptual and governance contributions

Three papers contribute no primary data but supply instruments used throughout this chapter.

Huang et al.[23] survey hallucination in LLMs. Although general-domain, it provides the definitional apparatus applied in Section 6: a layered taxonomy separating factuality hallucination (contradiction, fabrication) from faithfulness hallucination (instruction, context and logical inconsistency), with root causes mapped across data, training and inference stages and matched to detection and mitigation strategies. Residual hallucination surviving retrieval augmentation receives separate treatment, which bears on the retrieval-augmented systems examined in Section 3.

Chung et al.[10] map five architectural challenges, hallucination, interpretability, bias, privacy and clinical effectiveness, to practical mitigations, drawing on deployment experience. The intrinsic/extrinsic distinction is adapted to counselling dialogue: intrinsic hallucination contradicts the dialogue history, whereas extrinsic hallucination cannot be verified against any available input. Stochastic neural generation is paired with deterministic hard-coded overrides for crisis detection, on the premise that hallucination cannot be eliminated within the current paradigm and must instead be contained.

Asman et al.[2] frame a theme issue on responsible generative AI in mental health, offering a governance roadmap that integrates a relational ethics of care with clinical alliance standards, together with a proposed clinical risk assessment questionnaire for patients and developers.

1.4 Appraisal of the secondary evidence

Four properties recur across the seven empirical reviews and define the gap this work addresses. None performs quantitative synthesis, a defensible response to genuine heterogeneity given that the primary literature reports F1, balanced accuracy, AUC, BLEU, ROUGE, κ, ICC and bespoke Likert instruments without shared task definitions, but the consequence is that no pooled effect estimate exists for any LLM mental health application. Quality appraisal is the exception rather than the norm: two of seven apply a formal instrument and two report inter-reviewer agreement, so five synthesise studies of unassessed quality. Inclusion rates range from 0.27% to 10.9%, a fortyfold spread across overlapping periods and topics, more consistent with differing operational definitions of "LLM" and "mental health application" than with differing literatures.

Every search window closes before the majority of this corpus was published. The latest end-date among the seven is August 2024 and the earliest December 2023, whereas 45 of 66 studies here (68%) appeared in 2025 or 2026 (Table 1.2). The existing secondary literature therefore cannot speak to multi-turn adversarial safety benchmarking, agentic red teaming, LLM-as-judge reliability frameworks, or preference-optimisation approaches to alignment.

Table 1.2: Corpus publication years relative to prior review coverage

Year	**Studies in corpus**	**Share**	**Covered by any prior review's search window**
2023	4	6%	Yes
2024	17	26%	Partially
2025	36	55%	No
2026	9	14%	No
Total	**66**	**100%**	**32% at most**

A fifth observation concerns scope. Five of seven reviews organise around application areas. Only Yang et al., through its boundary framing, and Hua et al.[22], through its evaluation pyramid, organise around evaluative adequacy. None organises around methodology as such: how corpora are constructed, how models are adapted, how evaluation is designed, and how risk is assessed. That is the organising principle adopted here.

1.5 Positioning of the present review

Five commitments follow from the appraisal above. Extraction is methodology-only, so that where prior reviews chart what models achieved, this review charts how the achievement was established. Every study is extracted across the same twenty-four fields, including four risk fields (hallucination, bias, safety, human evaluation) that no prior review extracts systematically, which permits the corpus-wide statements made in Sections 6 and 7. The search window extends through 2026, capturing the 68% of the corpus outside all prior coverage. Absence is recorded as data, with "Not Reported" distinguished from "not applicable", so that gaps in the primary literature become findings. Two studies appeared in both preprint and published form and were merged into single records, giving 66 unique studies from 68 retrieved documents.

Two limitations should be stated at the outset. Like all seven reviews examined here, this review performs no quantitative pooling, for the same heterogeneity reason. No formal quality appraisal instrument is applied to included studies; methodological quality is instead characterised through the extraction fields themselves and appraised cross-sectionally in Section 7.

This is a weaker guarantee than a validated instrument would provide and represents a deliberate trade-off against extraction depth.

2 Data Foundations and Corpus Construction

Data constraints determine which methods are available. The recurring criticism that models are evaluated on social media proxies rather than clinical presentations is usually stated as a limitation, but read across twelve studies whose primary contribution is a corpus or an annotation protocol it is better understood as a structural condition. Clinical mental health data is scarce, legally encumbered and small, and most downstream choices examined in Sections 3 to 6 respond to that scarcity. Twelve studies (18.2%) are grouped here because their principal contribution is the construction, integration or annotation of a dataset rather than a modelling technique.

2.1 Clinical and interview corpora

Data generated inside a clinical or semi-clinical encounter underpins four studies, which with one exception are the smallest datasets in the corpus.

Ohse et al.[44] collected the KID corpus of German semi-clinical interviews, transcribed with Whisper large-v2. Fine-tuning on DAIC-WOZ and Extended DAIC makes the study simultaneously a corpus contribution and a generalisation test across language and protocol. Ground truth was PHQ-8 self-report, dichotomised at the conventional cut-off for classification and retained continuously for severity estimation. Acknowledged limitations include a unimodal text approach that discards prosodic and visual cues, translation-induced loss of nuance, and age and gender imbalance in recruitment.

Sood et al.[54] addressed scarcity by construction, integrating DAIC, E-DAIC and the Chinese-language EATD corpus into I-DAIC. Machine translation of EATD via the M2M 1.2B model was validated by bilingual annotators, the sole documented translation-verification step among the corpus-building studies. Five strategies for handling documents exceeding transformer context limits are benchmarked, from truncation variants to sliding windows and extractive summarisation. Fine-tuned BERT and RoBERTa underperformed TF-IDF-based SVM and logistic regression on weighted F1, attributed to class imbalance, which tempers any assumption that transformer adaptation is automatically superior on small clinical corpora.

Xu et al.[61] assembled the largest clinical dataset in this group: psychiatrist-patient dialogues from 1,160 outpatients at the Shanghai Mental Health Center, roughly 15,000 minutes of transcribed Mandarin audio with electronic medical records covering 25 diagnoses. Ground truth derives from psychiatrist documentation rather than self-report, validated against 50 EMR cases reviewed by specialist psychiatrists. The pipeline combines 138 validated clinical features and 177 scale items with an ensemble classifier over LLM outputs, LIWC counts, TF-IDF weights and OpenSMILE prosodic features, SMOTE oversampling, 10-fold cross-validation, Mann-Whitney U tests with FDR correction, and McNemar's test. Clinical noise degraded the acoustic features and Mandarin-centric training restricts transfer.

Xin et al.[60] used IMPACT-ME, 102 end-of-treatment interviews from 34 sets of depressed adolescents, parents and therapists nested within a randomised controlled trial. Fine-tuning was declined in favour of frozen embeddings with weighted classifiers because positive instances were too sparse, an explicit acknowledgment that method must follow data size. Cross-validation was grouped by subject ID to prevent leakage across triplets; no other study in this section documents subject-level leakage control. Some categories could not be trained at all.

2.2 Social media and platform corpora

Where corpora are drawn from public platform text, the variable that distinguishes them is label provenance rather than platform.

Qi et al.[47] introduced SOS-HL-1K (1,249 posts, suicide risk) and SocialCD-3K (3,407 posts, cognitive distortion), crawled from the Zoufan blog on Weibo. Three annotators with clinical psychology backgrounds worked double-blind, disagreements were resolved with a senior licensed psychologist, and agreement was quantified by both Fleiss' κ and Krippendorff's α. Supervised baselines were trained across five random seeds and LLMs evaluated over five runs at five temperature settings, with paired-samples t-tests at $\alpha = 0.05$. The reported advantage of supervised models over LLMs on complex clinical classification is supported by a documented annotation and testing protocol.

Nanda et al.[42] used the Self-reported Mental Health Diagnoses corpus, where labels derive from users' own disclosures on Reddit rather than clinical assessment. Construction is a sequence of heuristics: regex cleaning and token pruning, condition keyword lists from DSM-5 headings extended with synonyms, misspellings and vernacular terms, cosine-similarity filtering at 0.5, and downsampling of controls to approximately 8,000 posts. Each step is individually defensible, though in combination they determine which users enter the corpus. Evaluation rests on a single English-language dataset without clinical validation.

Sood et al.[53] created SMILE-College, 793 records of student narrative feedback from the College Pulse Student Voice Survey, annotated in two collaborative stages by a graduate student and a validator. A "Mixed" sentiment category emerged during annotation and was retained rather than collapsed. No formal agreement statistic is reported.

Table 2.1: Label provenance across the corpus-building studies

Provenance tier	**Mechanism**	**Studies**	**Agreement statistic reported**
Expert clinical annotation	Psychiatrist or psychologist coding of clinical or platform text	Xu et al.[61]; Qi et al.[47]; Roy et al.[48]	Fleiss' κ + Krippendorff's α (Qi); Cohen's κ 0.74/0.72 (Roy); psychiatrist-reviewed test set (Xu)
Structured self-report instrument	PHQ-8 or BDI-II administered under protocol	Ohse et al.[44]; Sood et al.[54]; Bucur[6]	Not applicable
Trained non-clinical annotators	Graduate students, bilingual validators, relevance raters	Sood et al.[53]; Bucur[6]	Majority-vote and unanimity gold standards (Bucur); none reported (Sood)
Platform self-disclosure	User-stated diagnosis matched by keyword patterns	Nanda et al.[42]	None
Model-generated pseudo-label	LLM assigns training labels at scale	Aggarwal et al.[1]; Hu et al.[20]	None (Aggarwal); human proofreading (Hu)

2.3 Synthetic and simulated corpora

Two studies generate rather than collect data, and both report fidelity problems that qualify their use as demonstrations.

De Duro et al.[12] produced CounseLLMe, 400 simulated counselling dialogues of 20 quips each, 200 English and 200 Italian, with a psychotherapist advising on prompt development. Evaluation reconstructs dialogues as Textual Forma Mentis Networks and profiles emotion using Plutchik lexicons, comparing against 24 human transcripts from HOPE with Kruskal-Wallis tests on network distances. Manual coding identified role-swapping and dialogue interference in Italian-language generation, and the models did not reproduce the anger and frustration characteristic of human patients.

Bucur[6] tested synthetic augmentation directly, generating 30 synthetic Reddit-style posts for each of 90 BDI-II responses (2,700 per model) using GPT-3.5 and Llama-3 8B, then testing whether these improved semantic retrieval from the eRisk 2023 corpus of roughly four million sentences. Relevance was established by top-k pooling ($k = 50$) with three annotators producing majority-voting and unanimity gold standards, evaluated by Average Precision, R-Precision, P@10 and NDCG@1000. Generated queries proved too specific and degraded retrieval relative to the original questionnaire responses. MPNet, fine-tuned for semantic search, outperformed MentalRoBERTa, which runs counter to the expectation that domain pre-training dominates.

2.4 Annotation protocols and label provenance

In three studies the production of labels, rather than the collection of text, is the object of investigation.

Roy et al.[48] built a diagnostic annotation layer over PRIMATE, prompting models to identify spans evidencing PHQ-9 and GAD-7 criteria with outputs constrained to structured formats. Ground truth was built in two stages: GPT-4o generated candidate annotations and three expert clinicians validated a subset, achieving Cohen's κ of 0.74 for PHQ-9 and 0.72 for GAD-7. The datasets were released. Fine-tuning proved difficult, with the base MentaLLaMA model failing zero-shot tasks by repeating its input verbatim.

Aggarwal et al.[1] interrogate label provenance directly, comparing four annotation sources on identical Reddit data, SMHD dictionary matching against Llama3, MentaLLaMA and Samantha-Mistral, then fine-tuning downstream classifiers on each. The LLM annotators exhibited severity inflation, with the majority of posts labelled Severe and one model producing no Mild labels. Classifiers trained on LLM pseudo-labels showed large train-validation gaps (95.8% against 70.31% for Llama3), which the authors attribute to overfitting on unreliable labels. Taken together, these results indicate that LLM-generated training labels were not a neutral substitute for expert annotation in the tasks examined.

Hu et al.[20] constructed a large Chinese psychological training corpus, 155k single-turn QA pairs, 11k multi-turn dialogues and 10k textbook knowledge items, using a three-step pipeline of generation, evidence judgment and refinement. The evidence-judgment stage extracts matching source segments for each generated sentence, providing a grounding check against unsupported generation, and human proofreaders validated the output. The authors note potential generation bias from using LLMs to compile supervised fine-tuning data, which is the concern Aggarwal et al. report empirically.

2.5 Appraisal of the data foundations

Clinical data is the exception: four of twelve studies use data generated in a clinical encounter, and only Xu et al.[61] exceeds a thousand participants, so claims about clinical performance rest on a narrow base. Agreement statistics are rarely reported, with three of twelve giving a formal inter-annotator statistic and the remaining nine using single annotators, unquantified multi-stage validation, or labels derived from self-report or model output. Where a corpus becomes a shared benchmark, unquantified annotation quality propagates into every study that uses it.

Class imbalance is endemic and unevenly handled, named as limiting by four studies, with treatment ranging from SMOTE oversampling and weighted loss with subject-grouped folds to nothing at all. Leakage control is largely undocumented, with only Xin et al. grouping cross-validation folds by subject. Since several corpora contain multiple documents per participant, reported performance may be inflated in ways the reported metrics would not reveal.

Table 2.2: Corpus scale and validation characteristics

Study	Corpus	Scale	Ground truth source	Imbalance handling	Leakage control
Ohse et al.[44]	KID (+ DAIC/E-DAIC)	Semi-clinical interviews	PHQ-8 self-report	Not reported	Not reported
Sood et al.[54]	I-DAIC	626 documents	PHQ-8 self-report	Acknowledged, unresolved	Not reported
Xu et al.[61]	SMHC dialogues	1,160 outpatients, ∼15,000 min	Psychiatrist EMR + 50-case expert test set	SMOTE + 10-fold CV	Not reported
Xin et al.[60]	IMPACT-ME	34 subjects, 102 interviews	Qualitative content coding	Weighted loss	Subject-grouped 4-fold CV
Qi et al.[47]	SOS-HL-1K; SocialCD-3K	1,249 + 3,407 posts	3 clinical annotators + adjudicator	Not reported	Not reported
Nanda et al.[42]	SMHD subset	∼8k control posts	Platform self-disclosure	Downsampling	Not reported
Sood et al.[53]	SMILE-College	793 records	2-stage student annotation	Not reported	Random 75/5/20 split
De Duro et al.[12]	CounseLLMe	400 dialogues	Synthetic (HOPE as reference)	Not applicable	Not applicable
Bucur[6]	eRisk 2023 + synthetic	∼4M sentences; 2,700 synthetic/model	3 annotators, majority + unanimity	Not applicable	Not applicable
Roy et al.[48]	PRIMATE + released sets	Randomised subsets	GPT-4o proposal + 3 clinician validation	Not reported	Not reported
Aggarwal et al.[1]	Reddit	5,000 posts	LLM pseudo-labels (4 sources compared)	5-fold CV	Not reported
Hu et al.[20]	Chinese corpus	155k + 11k + 10k items	LLM generation + evidence judgment + proofreading	Not reported	Not reported

Both studies that generated training or query data report fidelity failures, and the study that used model-generated labels at scale documented systematic severity inflation. Against this, Hu et al. show that an explicit evidence-judgment stage can impose grounding discipline on generated data. Across these studies, synthetic data proved usable only where a verification stage was built in, which argues for treating verification as a default rather than an optional addition.

3 Model Adaptation Paradigms

Fourteen studies (21.2%) contribute a technique for adapting a general-purpose language model to a mental health task rather than a corpus, benchmark or clinical evaluation. Four strategies are represented, best understood as positions on a trade-off between the cost of changing model weights and the cost of controlling behaviour at inference. Prompt engineering changes nothing and controls at inference; fine-tuning changes weights and accepts the compute and data cost; retrieval augmentation leaves weights untouched but constrains generation against an external evidence store; orchestration composes several models or tools without adapting any. A fifth, smaller line abandons generation and treats the model as a feature extractor.

3.1 Prompt engineering and structured reasoning

Patil and Gedhu[46] compared Chain-of-Thought, Self-Consistency CoT, Few-Shot CoT and Tree-of-Thought on a single reasoning-oriented model (o3-mini) across five Reddit benchmarks. Holding the model constant while varying only the reasoning scaffold isolates the effect of the strategy, which multi-model comparisons cannot. The metric suite extends beyond accuracy and F1 to Matthews Correlation Coefficient, Quadratic Weighted Kappa, Mean Absolute Error and both ROC and PR AUC, appropriate choices for imbalanced ordinal tasks. Few-Shot CoT performed best on multi-class tasks, plain CoT and SC-CoT on binary tasks, and Tree-of-Thought was inconsistent throughout. No inferential statistics are reported, so no claim of significant difference is supported. Performance degrades in multi-class and long-text settings and remains sensitive to minor phrasing variation, which may indicate fragility in prompt-based adaptation.

Kim et al.[29] developed LLM4CBT, a training-free prompt-alignment framework combining therapist persona, CBT technique examples and preferred clinical behaviours. Evaluation covers real therapist utterances from a merged HighQuality and HOPE corpus (366 dialogues, 7,669 utterances) against ground-truth act labels across 13 act types, plus simulated conversations in which elicited automatic thoughts are checked against ground truth embedded in generated patient profiles. Testing pacing against deliberately reluctant patients is a useful stress condition. A circularity risk is acknowledged, since the evaluation pipeline depends on automated LLM classifiers and so shares the failure modes of the system measured.

Xu et al.[62] span this subsection and the next, and are placed here on the basis of their prompt taxonomy. The zero-shot template decomposes into four components, with three enhancement strategies compared against a basic condition: context enhancement, mental health enhancement, and their combination. Few-shot and chain-of-thought variants extend the design, and crossing this axis with instruction fine-tuning allows the prompting-versus-tuning trade-off to be quantified on identical tasks, which no other study in the corpus does. Compact fine-tuned models (Mental-Alpaca 7B, Mental-FLAN-T5 11B) match task-specific Mental-RoBERTa and exceed GPT-4 in zero- and few-shot settings, which suggests that prompt-based adaptation of large proprietary models is not the efficient frontier for classification.

3.2 Supervised and parameter-efficient fine-tuning

Five studies update model weights. The compute range they span bears on which research groups can undertake this work.

Lai et al.[31] built Psy-LLM through domain pre-training on 2.85 GB of crawled Chinese psychology data over 100,000 iterations, then fine-tuning on 56,000 PsyQA pairs, on a single V100. Evaluation is dual: intrinsic metrics (perplexity, ROUGE-L, Distinct-1/2) alongside human evaluation in which six psychology-faculty students scored 200 pairs on helpfulness, fluency, relevance and logic. Outputs are conceded to fall short of human quality, and the unidirectional architecture constrains contextual memory.

Maurya et al.[37] compared four lightweight sequence-to-sequence models fine-tuned on 20,500 QA pairs from four public counselling resources, with a 70/30 split on a single T4. Reporting hardware metrics alongside quality metrics makes deployability an explicit evaluation axis. The quality metrics are weaker: ROUGE, BLEU-4 and perplexity measure n-gram overlap and are insensitive to empathy or clinical appropriateness, and the human assessment was conducted by a single non-expert evaluator.

George et al.[14] fine-tuned Mistral-7B Instruct v0.1 with QLoRA on the counsel-chat dataset using a single RTX 3090. Evaluation departs from n-gram metrics in favour of PsychoBench, assessing Emotional Intelligence Scale and Empathy Scale scores against human reference averages from prior literature, with F-tests and t-tests. Applying instruments designed for humans to a model is of contested construct validity, though it aligns more closely with counselling quality than n-gram overlap.

Gumilang et al.[15] pursued cultural rather than architectural adaptation through a three-stage pipeline: global pre-training on Counsel Chat, expert-guided retraining on 1,800 locally sourced Indonesian entries, and L2 regularisation to counter overfitting on the small local corpus. Evaluation combines training and validation accuracy tracking with User Acceptance Testing among 58 students on Likert-scaled ease of use, responsiveness and satisfaction. Accuracy curves and satisfaction ratings jointly say little about clinical quality, and the system has no crisis detection or intervention protocol.

Bookanakere et al.[5] fine-tuned GPT-2 (1.5B) quantised to 4 bits with a documented configuration chosen to fit strict resource constraints. The pipeline transforms tabular psychiatric lifestyle data into narrative paragraphs before training, evaluated on a held-out partition of 1,000 cases. Two conceded limitations are material: quantisation entails precision loss, and the source dataset is synthetic, so tabular-to-text transformation may not resemble clinical language.

Table 3.1: Compute footprint and evaluation instrument across fine-tuning studies

Study	Base model	Method	Hardware	Primary evaluation	Human evaluation
Lai et al.[31]	PanGu 350M / WenZhong 110M	Domain pre-training + SFT	V100 32GB	Perplexity, ROUGE-L, Distinct-1/2	6 psychology students, 200 pairs
Xu et al.[62]	Alpaca 7B / FLAN-T5 11B	Multi-task instruction FT	8 × A100 80GB	Balanced accuracy, cross-dataset transfer	Not reported
George et al.[14]	Mistral-7B Instruct	QLoRA	1 × RTX 3090 24GB	PsychoBench EIS/Empathy, F-test, t-test	Human norms from literature
Maurya et al.[37]	T5, FLAN-T5, BART, GODEL (60–220M)	SFT	1 × T4 16GB	ROUGE, BLEU-4, perplexity + hardware metrics	1 non-expert evaluator
Gumilang et al.[15]	GPT-4-based	SFT + L2 regularisation	Not reported	Train/validation accuracy	58 students (UAT, Likert)
Bookanakere et al.[5]	GPT-2 1.5B	4-bit quantised FT	Edge-constrained	Accuracy, precision, recall, F1	Not reported

3.3 Retrieval-augmented and knowledge-grounded architectures

Retrieval-based designs leave weights untouched and constrain generation against external evidence. All three invoke hallucination reduction as motivation, though they differ in whether they substantiate it.

Kharitonova et al.[27] built a retrieval system over the Clinical Practice Guidelines of the Spanish National Health System, using semantic search on paragraph embeddings with a negative-constraint prompt confining answers to retrieved content, isolating reasoning ability from parametric knowledge. Twenty expert-authored questions were rated by a specialist clinician on three Likert scales with independent verification by a second medical expert. The Veracity scale, scored from -1 to $+1$, penalises non-answers, so a system cannot score well by refusing to respond, a failure mode that abstention-based designs invite. The claim that hallucination was eliminated by the closed-database design follows from architecture rather than measurement. Open-source LLaMA models answered more often and more accurately than GPT-3, which declined 25% of ADHD questions.

Guo and Guo[16] integrated GraphRAG over a knowledge graph constructed from DSM-5 and ICD-11 criteria with EmoLLM for symptom extraction, requiring no additional training. Consultation dialogues are segmented into k-round groups ($k = 6$), candidate symptoms extracted, then validated and deduplicated against the knowledge graph before a CoT-based diagnostic step. Rather than relying on retrieval to suppress fabrication implicitly, each extracted symptom is re-checked against a structured clinical ontology. The evaluation is appropriately ablated across zero-shot model comparison, backbone substitution, segmentation length and component removal, though it rests on a single small dataset (MMDA, 524 samples).

Bhanu Sree et al.[4] combined retrieval augmentation with LangChain agents executing specialised tools in parallel, routing queries by source type. The evaluation is almost entirely operational: time to first token, tokens per second and latency per minute, with qualitative feedback on conciseness, contextual accuracy, helpfulness and relevance. Latency outliers up to 495 seconds are reported. No hallucination measurement is reported despite grounding in PubMed being the stated rationale, and general pre-trained models are used without domain-specific updates.

3.4 Orchestration and hybrid pipelines

Two studies compose components without adapting any of them, leaving the assembled system largely unevaluated.

Kamoji et al.[26] built a two-stage pipeline in which an ensemble classifier predicts mental health status from a 41,000-entry survey dataset with 25 features, after which Gemini generates personalised natural-language insight. Keeping the diagnostic decision with an interpretable model constrains the LLM to explanation. Two features limit what can be concluded. Reported accuracy near 99% on self-report data is, as the authors concede, suggestive of overfitting. The LLM component is not evaluated for insight quality, factual accuracy or safety, despite the reports being intended for psychiatrist review.

Singh et al.[51] documented MindGuide, a LangChain composition using LLMChain with ConversationBufferMemory over GPT-4 at temperature 0.5 through a Streamlit interface. The architecture is documented in reproducible detail. Evaluation consists of a qualitative walk-through of four conversation stages, with no metrics, dataset, comparison condition or formal human evaluation, which the authors state as a limitation. The system is nonetheless positioned as an early identification assistant for anxiety, depression and suicidal thoughts, with no crisis escalation protocol, guardrail or safety evaluation reported.

3.5 Embedding-based adaptation

Luna-Jiménez et al.[36] abandon generation and use the LLM as a frozen feature extractor. Last-layer 4,096-dimensional d-vectors from Llama-2-7b-chat and MentaLLaMA-chat-7B were averaged temporally, ranked by F-ANOVA filter and passed to conventional classifiers. Evaluation used 80/10/10 splits on Counsel-Chat and 7Cups with weighted F1 alongside 95% confidence intervals, one of the few studies here to quantify uncertainty in its headline metric. Roughly 75% of dimensions proved redundant, permitting large reductions in downstream cost, though selected dimensions show low overlap across datasets, indicating a dataset-specific rather than transferable ranking.

3.6 Appraisal of the adaptation literature

Evaluation depth stands in inverse relation to engineering scope. Studies presenting full pipelines, agentic orchestration or deployed interfaces report the least evaluation: Singh et al. report no metrics, Kamoji et al. do not evaluate the generative component of a two-stage system, and Bhanu Sree et al. measure latency but not accuracy or hallucination. Narrower studies report more, with Patil and Gedhu giving ten metrics across five datasets, Luna-Jiménez et al. giving confidence intervals, and Kharitonova et al. designing a scoring scale to close an abstention loophole. Architectural novelty and evidential strength vary largely independently in this literature.

Generation quality is measured with instruments known to be inadequate. Four studies generating therapeutic text rely wholly or partly on ROUGE, BLEU or perplexity, which measure overlap with a reference and are insensitive to empathy, clinical appropriateness or harm. Alternatives are available, including PsychoBench scales, act-label distribution matching against human therapists, and expert Likert rating with a non-answer penalty, but none has been widely adopted.

Human evaluation, where present, is limited, ranging from six psychology students rating 200 pairs down to a single non-expert evaluator, with five studies reporting none. No study here reports inter-rater agreement. Hallucination control is asserted architecturally more often than measured: all three retrieval studies invoke hallucination reduction, but one implements an explicit verification stage, one argues from closed-database design without measuring, and one measures nothing. Section 6 distinguishes hallucination mitigation from hallucination evaluation on this basis.

Compute access shapes the method distribution. Three studies frame parameter-efficient or quantised adaptation as enabling deployment in low-resource settings, where clinician shortages are most acute. The more deployable systems are trained on the smallest and least curated corpora, compounding the data-quality concerns raised in Section 2.

Table 3.2: Evaluation adequacy across the adaptation studies

Adaptation strategy	Studies	Quantitative task metrics	Generation-quality instrument	Human evaluation	Hallucination measured
Prompting	Patil, Kim, Xu	All three	Act-label match (Kim)	LLM-based (Kim)	No
Fine-tuning	Lai, Maurya, George, Gumilang, Bookanakere	All five	ROUGE/BLEU (Lai, Maurya); PsychoBench (George)	4 of 5, quality varies	No
Retrieval-augmented	Kharitonova, Guo, Bhanu Sree	Guo only	Expert Likert (Kharitonova)	Kharitonova (2 experts); Bhanu Sree (informal)	Verification stage (Guo)
Orchestration	Kamoji, Singh	Kamoji (classifier only)	None	None	No
Embeddings	Luna-Jiménez	Yes, with 95% CIs	Not applicable	No	Not applicable

4 Benchmark Construction and Automated Evaluation

Six studies (9.1%) contribute an evaluation artefact: a benchmark, a scoring protocol, or a framework for deciding when automated evaluation can be trusted. They are treated before human and clinical evaluation because the recurring question in that literature is whether expert raters are necessary, and that question is answerable only once the automated alternative and its documented failure points have been characterised. Two of the six address it directly by evaluating LLM judges against human experts on identical material. Three groupings emerge, distinguished by the source of ground truth: examination-grounded benchmarks inherit it from professional licensing instruments, expert-anchored benchmarks generate it from clinician panels, and judge-reliability studies treat the relationship between the two as the object of measurement. Methodological rigour in this corpus is concentrated here: five of six report inferential statistics, three report formal reliability coefficients, and two quantify uncertainty in their headline estimates.

4.1 Examination-grounded competency benchmarks

Nguyen et al.[43] constructed CounselingBench from professional licensing standards: 1,612 multiple-choice questions across 138 clinical case studies with expert-written rationales. Twenty-two models were crossed with five core competencies under zero-shot, few-shot, chain-of-thought and self-consistency decoding. Pairing each medical-specialised model with its own general-purpose counterpart isolates the effect of domain fine-tuning from architecture, so the shortfall of 4.2 percentage points relative to generalist bases can be attributed to fine-tuning rather than to model family. Reasoning errors were manually annotated on 100 responses by three expert annotators. Competency mapping used two psychiatrists with a licensed counsellor as tiebreaker, and paired t-tests and chi-square tests support the accuracy and error-distribution claims. Multiple-choice cannot capture empathy or therapeutic alliance, and a US licensing exam has limited applicability to the Global South.

Xu et al.[63] built the first integrated Chinese benchmark, combining the Chinese Counselor Qualification Exam (744 single-choice and 200 multiple-choice items, 2023–2024) with translated social media diagnostic screening on Dreaddit (1,151 posts) and SDCNL (1,517 posts). The two-task structure separates theoretical knowledge from applied diagnostic performance. The two appear to dissociate, since models strong on examination content are not correspondingly strong on classification. Fifteen models spanning 1.5B to 671B parameters were queried through a programmatic API harness at default parameters, a choice that maximises reproducibility while not exploring decoding sensitivity. Class balance was maintained in SDCNL test sets, and models over-predict the anxiety class under uncertainty. Analysis is descriptive, with accuracy compared across parameter scales and release timelines and no inferential testing. Ground truth quality is uneven across sources, with professional examiners for the exam but crowdworkers for the two social media corpora.

Hanss et al.[18] took 150 single-answer multiple-choice questions from a psychiatry review manual and administered each ten times to GPT-3.5, GPT-4 and GPT-4o, allowing up to fifteen attempts to obtain ten valid responses. Rather than reporting a single accuracy figure, response consistency is computed via frequency variance across trials and tested as a predictor of correctness, alongside separately elicited pre-answer self-reported confidence. Consistency predicts accuracy while self-reported confidence does not, which offers a proxy for reliability requiring no ground truth at inference time. Analysis used chi-square tests under Bonferroni correction at $\alpha = .01$, t-tests for consistency and point-biserial correlations for predictors, with decoding parameters documented. Two threats are acknowledged: the format cannot assess interactive dialogue, and the questions may already sit in model training data, a contamination risk intrinsic to benchmarks built from published materials.

Table 4.1: Examination-grounded benchmark parameters

Study	Instrument	Scale	Models	Repetition	Inferential statistics
Nguyen et al.[43]	Professional licensing standards	1,612 MCQs, 138 case studies	22 (13 medical + matched generalists)	4 prompting/decoding conditions	Paired t-tests; chi-square
Xu et al.[63]	Chinese Counselor Qualification Exam + 2 social corpora	944 exam items; 2,668 posts	15 (1.5B–671B)	Single pass, default parameters	None (descriptive)
Hanss et al.[18]	Psychiatry review manual	150 MCQs × 10 trials	3 (GPT family)	10 trials per item	Chi-square (Bonferroni, α=.01); t-tests; point-biserial r

4.2 Expert-anchored benchmarks

Two studies generate ground truth from clinicians rather than inheriting it, which permits assessment of open-ended responses.

Li et al.[34] assembled the largest clinical expert panel among the included studies. COUNSELBENCH-EVAL comprises 100 real patient questions from CounselChat with four responses each, three LLM-generated and one from a verified human therapist, rated blind by 100 licensed or professionally trained practitioners. Rating covers six dimensions with scales matched to the construct, and raters extracted supporting text evidence and wrote rationales, so judgements trace to specific spans rather than holistic impressions. Inter-rater reliability is reported as Krippendorff's α, appropriate for the mixed scale types. Inference uses Wilcoxon signed-rank tests, with Kruskal-Wallis and chi-square examining whether rater experience or time-on-task influenced ratings; few other studies here report rater-effect control. A second, adversarial component had ten clinicians author 120 questions targeting six failure modes identified in the first phase, with five different practitioners evaluating 1,080 responses, which prevents self-marking. Hallucination is operationalised as Factual Consistency scoring validated against expert annotation, and safety as Medical Advice and Toxicity flags. A pilot excluded three domain-tuned models for high invalid-response rates, which is itself a finding. Interaction is restricted to a single turn, and the design depends on scarce publicly available non-deidentified data.

McBain et al.[38] repurposed the Suicide Intervention Response Inventory (SIRI-2), 24 hypothetical patient remarks paired with two clinician responses each, as an LLM benchmark. Prompting was deliberately left unengineered, using original SIRI-2 instructions only and zero-shot, to isolate baseline competency rather than optimised performance. Because SIRI-2 carries published norms, model scores can be positioned on a human scale spanning expert suicidologists, clinical PhDs, master's-level counsellors and untrained K-12 school staff, which is more interpretable than an accuracy percentage. Three independent research accounts prompted each model, supporting test-retest correlation. Analysis accounts for nested item structure through linear regression with z-score outlier detection at $|z| > 1.96$, conducted under STROBE reporting. The stated limitation is precise: the study measures evaluative competency, the model as referee, not active engagement with a person in crisis.

4.3 Judge reliability and the limits of automated evaluation

Badawi et al.[3] address when an LLM judge can substitute for a human expert. MentalBench-100k comprises 10,000 conversations from three sources and MentalAlign-70k comprises 70,000 ratings. From 1,000 curated contexts, ten responses each were generated, one human and nine model, and rated by four LLM judges and three human experts with graduate-level or licensed psychiatric backgrounds across seven attributes grouped into a Cognitive Support Score and an Affective Resonance Score.

Agreement is decomposed into consistency ICC(C,1) and absolute agreement ICC(A,1). The distinction is consequential because a judge can rank responses correctly while remaining systematically miscalibrated, and only the second coefficient detects this. Systematic bias is reported directly as the LLM-human difference. Precision is quantified through 95% bootstrap confidence interval widths from 1,000 nonparametric iterations, with two-way mixed-effects ANOVA variance decomposition and paired t-tests supporting inference across 28 judge-attribute combinations. Ratings were aggregated to model level to filter conversation-level noise, and traditional error metrics serve as baselines. The resulting Affective-Cognitive Agreement Framework is conditional rather than categorical, classifying where automated judgement appears reliable and where human oversight remains necessary, which reframes automated evaluation as a calibration problem. Coverage is English-only and single-turn, running multiple large judges carries a computational cost, and three raters supported 70,000 ratings.

4.4 Appraisal of the benchmark literature

Statistical reporting is stronger here than elsewhere: five of six studies report inferential statistics, against roughly one in five across the adaptation literature, three report formal reliability coefficients and two quantify uncertainty. Xu et al.[63] reports descriptive accuracy only, despite the largest model sweep in the subsection.

Stochasticity is taken seriously, but unevenly: Hanss et al. administer ten trials per item and convert the variance into a reliability measure, McBain et al.[38] use three independent instances, Li et al. fix and document decoding parameters, and Xu et al.[63] run a single pass at default settings. Since identical prompts yield different answers across runs, single-pass results carry unreported variance, and no repetition standard has been established.

The multiple-choice format is the binding constraint on examination-grounded work, since performance cannot capture empathy, therapeutic alliance or interactive competence. Expert-anchored designs recover open-ended assessment at a cost visible in their sample sizes, 100 questions and 48 items respectively, and the trade-off between construct validity and scale remains unresolved. Contamination is acknowledged once and otherwise unaddressed, though the risk applies to every benchmark built from published instruments.

Domain fine-tuning does not help, on two independent measurements. Nguyen et al. find medical-specialised models underperform matched generalist bases by 4.2 percentage points, and

Li et al. excluded three domain-tuned models for high invalid-response rates. Combined with the finding in Section 3 that compact instruction-tuned models outperform far larger prompted ones, this suggests that within the reviewed studies domain-directed fine-tuning was not consistently associated with improved accuracy, whereas task-directed adaptation was.

Table 4.2: Evaluation rigour across the benchmark studies

Study	Ground truth source	Reliability statistic	Uncertainty quantified	Repetition	Rater-effect control
Nguyen et al.[43]	Licensing exam key + 2 psychiatrists	Tiebreaker adjudication	No	Prompting conditions	Not reported
Xu et al.[63]	Exam key + crowdworker labels	None	No	Single pass	Not applicable
Hanss et al.[18]	Published review manual key	Response consistency (variance)	Bonferroni α =.01	10 trials/item	Not applicable
Li et al.[34]	100 licensed practitioners, blinded	Krippendorff's α	No	Fixed decoding	Kruskal-Wallis/chi-square on experience and time
McBain et al.[38]	Expert suicidologist norms + historical cohorts	Test-retest correlation	z-score outlier detection	3 instances	Nested-structure regression
Badawi et al.[3]	3 psychiatric experts, 70k ratings	ICC(C,1) and ICC(A,1)	95% bootstrap CIs (1,000 iters)	4 judges × 7 attributes	Two-way mixed-effects ANOVA

Two studies test whether LLM judges align with human experts on the same material, quantifying the conditions under which automated judgement holds rather than assuming it. The distinction Badawi et al. draw between consistency and absolute agreement applies equally to the human rating protocols examined in Section 5, and is taken up again in Section 7.

5 Human-Centred and Clinical Evaluation Designs

Expert time constrains most of the designs examined below. Fourteen studies (21.2%) place human judgement or human experience at the centre of evaluation. They are ordered here along a validity ladder. Vignette studies maximise experimental control at the cost of ecological realism; expert-rater studies apply clinical judgement to real model outputs; controlled trials and prospective deployments observe systems in use; qualitative work studies naturalistic interaction without intervening; and population survey evidence describes what is happening at scale. Internal validity is highest at the top of this ladder and ecological validity at the bottom, and no study achieves both.

5.1 Vignette and standardised-case designs

Controlled clinical stimuli are used in four studies, which multiply small vignette sets through repetition to reach analysable sample sizes.

Levkovich[33] administered six published clinical vignettes to four LLMs twenty times each in parallel male and female versions, giving 960 evaluations. Comparison norms were inherited from the source studies, covering 1,536 health professionals and 6,016 members of the public. Open-ended diagnoses were scored by automated string-search filters, with chi-square and Fisher's exact tests and Cramér's V effect sizes. The gender-balanced design also functions as a bias probe. Claims are qualified by the use of hypothetical vignettes, the absence of multi-turn dynamics, and the lack of clinical validation.

Schnepper et al.[49] used a randomised 2 × 2 design crossing gender with sexual orientation across 30 published eating-disorder vignettes, yielding 120 variants evaluated in three rounds by each of two models, for 720 evaluations. Outcome instruments are validated psychometrics rather than ad-hoc scales, the RAND-36 Mental Composite Summary and the EDE-Q global score, which renders any bias effect interpretable in clinically familiar units. Analysis uses linear multilevel models with crossed random effects and intraclass correlation coefficients testing replicability, with temperature set to 0. Three limitations apply: the source vignettes were gender-imbalanced, the custom temperature controls are unverified, and MentaLLaMA produced data of sufficiently low quality that the general-versus-specialised comparison is weakened.

Thotapalli et al.[57] evaluated three GPT variants across 22 youth psychiatric emergency vignettes over four iterations (264 responses) against two advanced practice nurse practitioners, with three board-certified psychiatrists grading response quality. Agreement was quantified with quadratically weighted Cohen's κ for clinician-LLM comparison, appropriate for ordinal triage categories where the distance between disagreements matters, alongside Fleiss's κ for inter-model consistency. An error asymmetry was observed, with zero false negatives against up to four false positive admissions, which is the safety-preferable direction but carries resource implications. Vignettes were drafted using ChatGPT and refined by the research team, and the resulting circularity should be weighed when interpreting the results.

Lauderdale et al.[32] ran three models across four standardised veteran vignettes over ten trials each, scored against the Veterans Health Administration Risk Stratification Table and compared with 42 licensed mental health care providers. Scoring used an operational structured professional judgement framework already in clinical use, with acute risk, chronic risk and treatment disposition rated separately. Most other studies collapse the acute/chronic distinction. Analysis used independent-groups t-tests, one-way ANOVA with Tukey's HSD and hierarchical regression. Four vignettes cannot support counterbalancing of demographic factors, and existing VHA machine-learning risk models were validated largely on White male veterans.

Table 5.1: Vignette study designs

Study	Vignettes	Repetition	Total observations	Human comparator	Reliability statistic
Levkovich[33]	6 (× male/female)	20 per version	960	1,536 professionals + 6,016 public (published norms)	Cramér's V effect sizes
Schnepper et al.[49]	30 → 120 variants (2×2)	3 rounds	720	None (published literature source)	ICC for replicability
Thotapalli et al.[57]	22	4 iterations	264	2 nurse practitioners + 3 psychiatrists	Weighted Cohen's κ; Fleiss's κ
Lauderdale et al.[32]	4	10 trials	120	42 licensed providers	Not reported

5.2 Expert-rater comparative studies

Settanni et al.[50] evaluated 425 Italian-language Reddit posts, filtered from 4,462, across three models under two prompt conditions, producing 2,550 ratings against a gold standard from two licensed clinical psychologists. Models output only a numerical urgency score on an adapted Mental Health Triage Scale customised for informal social media register. The evaluation reports continuous agreement alongside binary classification metrics after dichotomising at the urgent threshold, with repeated stratified 5-fold cross-validation over ten repetitions and temperature fixed at 0. Weighted Cohen's κ quantifies clinician inter-rater reliability. As in Thotapalli et al., over-triage is the dominant error, which the authors identify as the safety-relevant failure mode.

Cui et al.[11] developed a suicide intervention chatbot around a five-stage crisis protocol grounded in intervention manuals and the ACT model, deployed via a web platform with documented decoding parameters. Twenty psychology professionals conducted guided therapy sessions and completed an eight-statement questionnaire across six dimensions on 7-point Likert scales, including a dedicated Safety and Privacy dimension. Analysis is descriptive, with no comparison condition and no inter-rater statistic, which limits the design to establishing expert acceptability rather than efficacy.

Eberhardt et al.[13] reports the fullest psychometric validation in the corpus. From 1,131 video-recorded therapy sessions across 155 outpatients it develops an automated rating scale for latent patient engagement through a full psychometric pipeline, with 120 candidate items pre-selected on distributional properties, factor loadings and item-total correlations, then validated against independent process and outcome scales. Reliability is Cronbach's $\alpha = 0.952$, McDonald's $\omega = 0.953$ and average variance extracted $= 0.715$, with multilevel confirmatory factor analysis giving CFI $= 0.968$, TLI $= 0.956$, SRMR $= 0.022$ and RMSEA $= 0.108$, the last exceeding conventional thresholds and reported rather than suppressed. Overfitting is addressed through 1,000-resample bootstrap optimism correction and 3-fold cross-validation. Transcription quality is quantified, which few other studies using transcribed speech report. The model runs locally on secure institutional servers, so patient data does not leave the institution. Processing cost is high and the patient-level sample of 155 is modest for scale validation.

5.3 Controlled trials and prospective deployment

Campellone et al.[9] reports the only randomised controlled trial in the corpus: 160 adults randomised 1:1 to a generative AI arm ($n = 81$) or an identical rules-based comparator ($n = 79$) over two weeks, double-blind and decentralised. Guardrails operate at four tiers: a classifier that blocks concerning language and routes to helplines before any model call; an embedding-based off-topic classifier; prompt-level clinical constraints; and output-level structure and banned-word checks. Readiness was tested pre-trial against 42 simulated patient personas, and all 2,207 generated completions were manually annotated post-trial by two coders with principal-investigator resolution, verifying guardrail success exhaustively rather than by sampling. Analysis is descriptive, since the study was explicitly not powered for between-group inference and did not adjust for baseline severity.

Patias et al.[45] describe the me_HeLi-D project, integrating GPT-4 into digital modules for adolescents aged 12–15 covering early screening from journal entries, CBT and mindfulness interventions, interactive lessons and conversational chatbots. Planned evaluation is mixed-methods, combining pre/post surveys using the Mental Health Literacy Scale and Strengths and Difficulties Questionnaire with interaction logs, focus groups and behavioural tracking. Data protection is specified concretely. The study is ongoing and its conclusions rest on preliminary data, so it contributes a protocol rather than findings. It is included as the only prospective work with minors in the corpus. Automated screening of adolescent journal entries raises consent questions that the paper acknowledges but does not resolve.

5.4 Qualitative, ethnographic and naturalistic studies

Four studies observe interaction rather than staging it, and supply what evidence the corpus contains on system behaviour outside a research protocol.

Wang et al.[59] structured evaluation in three stages with deliberate separation of roles: co-design with five mental health professionals averaging fifteen years of experience, a field user study with 19 participants producing 95 transcripts, and transcript review by four different professionals who had not participated in the design. Reviewers used a think-aloud protocol over randomised dialogue logs, analysed thematically with iterative consensus-building. Separating designers from evaluators prevents self-marking, a control absent from most system papers in Section 3. Thematic codes name specific clinical failure modes including toxic positivity, leading questions and insufficient Socratic depth.

Song et al.[52] conducted semi-structured interviews with 21 participants recruited purposively and by snowball sampling from specialised online communities, with deliberate diversity across Indian, Nigerian, Brazilian and Central Asian participants. The sample is the most geographically diverse in the corpus. Since general-purpose models carry embedded linguistic and cultural bias, a Western-only sample would be unlikely to surface cultural mismatch. The analysis formalises therapeutic alignment as a multidimensional construct for assessing whether human-AI interaction supports long-term psychological healing. The sample is small and non-clinical, accounts are retrospective, and no clinical outcomes were measured.

Iftikhar et al.[24] ran an 18-month ethnography combining 110 self-counselling sessions conducted by seven trained peer counsellors, who met in 60 weekly focus groups to evaluate LLM behaviour, with 27 simulated multi-turn sessions independently and blindly reviewed by three licensed clinical psychologists. No other study in the corpus adopts a longitudinal design of this length. Inductive coding refined 41 initial codes to 15 ethical risks across five themes, each mapped to formal professional standards rather than to author-defined criteria. No quantitative analysis is reported, and the study covers only prompted rather than fine-tuned models on a single CBT-focused platform, with all psychologists from one country.

Moore et al.[41] analysed real chat logs from 19 users reporting psychological harm, 391,562 messages across 4,761 conversations. The pipeline is technically careful: transcripts de-identified with Presidio and Faker, tree-structured chat histories linearised into ancestral chains, and 28 codes across five categories applied by an LLM annotator with confidence thresholds raised for high-risk codes. Validation is layered, with a random 560-message sample annotated by seven co-authors including a board-certified psychiatrist, plus full manual verification of every suicidality and violence code. Agreement is reported as Cohen's κ for LLM-human and Fleiss' κ for inter-annotator agreement. Conversational dynamics are modelled through per-message regressions on log-transformed remaining conversation length with participant-clustered standard errors. The study is candid that agreement is moderate and prone to false positives on subjective codes, the sample is small and self-selected, and no objective clinical outcomes exist.

5.5 Population survey evidence

Stade et al.[55] surveyed 1,871 US adults between August and October 2024 using stratified sampling across age, sex and race/ethnicity to approximate national demographics, with hypotheses preregistered. All respondents answered demographic, psychopathology and treatment-use items; self-identified users additionally answered questions on how, why and for what they use LLMs, plus perceptions of usefulness, therapeutic alliance and the importance of memory. The estimate that 24% of respondents use LLMs for mental health, extrapolating conservatively to 14–18 million US adults, is the corpus's only population-scale figure, and its linkage to treatment access barriers supplies demand-side context absent from model-evaluation studies. Constraints follow from the design: cross-sectional self-report supports no causal inference, coverage is single-country and single-window, and screening measures substitute for diagnoses. This study's extraction is derived from optical character recognition, since the source document carries no text layer, and should be verified against the original before citation.

5.6 Appraisal of the human evaluation literature

Reliability reporting is stronger here than in other sections, with six of fourteen studies giving a formal agreement or reliability coefficient against three of twelve in Section 2 and none in Section 3, which is consistent with measurement discipline tracking proximity to clinical practice. Sample size tracks expert availability rather than the research question: two clinicians establish a gold standard for 2,550 ratings, three psychologists review 27 sessions, and four reviewers cover 95 transcripts. Vignette studies work around this by inheriting published norms, which is efficient but ties findings to whatever the source studies measured. Repetition is standard here, with all four vignette studies administering multiple trials and two fixing temperature at 0, so the stochasticity problem left unresolved in Section 4 is routinely handled.

The prospective evidence base is one underpowered trial: two weeks, $n = 160$, explicitly not powered for between-group inference, run by the product developer, and excluding participants with recent suicidality, precisely the population whose safety is most at issue. No study in this corpus reports clinical outcomes from a powered trial, leaving every efficacy claim dependent on proxy measures. Two triage studies independently report models erring toward escalation. This is the safety-preferable direction, and it contrasts with Section 6, where studies probing crisis handling find escalation too slow or absent.

Table 5.2: Position on the validity ladder

Design	Studies	Internal validity	Ecological validity	Reliability statistic reported
Vignette / standardised case	Levkovich, Schnepper, Thotapalli, Lauderdale	High (controlled stimuli, repetition)	Low (hypothetical, single-turn)	3 of 4
Expert rating of real outputs	Settanni, Cui, Eberhardt	Moderate–high	Moderate	2 of 3
Controlled trial / deployment	Campellone, Patias	Moderate (randomised but underpowered)	High	0 of 2
Qualitative / naturalistic	Wang, Song, Iftikhar, Moore	Low (no control condition)	Highest (real use, real harm)	1 of 4
Population survey	Stade	Not applicable (descriptive)	High (national sample)	Not applicable

Studies with the greatest ecological validity are those least able to support causal claims, while the study with the fullest psychometric validation measures a single construct on a single sample. No design in this corpus combines clinical realism with inferential power.

6 Risk, Safety and Assurance Evaluation

Assurance evaluates everything preceding it. A corpus can encode bias, an adaptation method can introduce fabrication, a benchmark can miss the failure that matters, and a clinical evaluation can be underpowered to detect harm. Ten studies (15.2%) treat those failures as the object of measurement. The section has a dual structure: ten studies have risk evaluation as their primary design and are treated individually, but risk fields are populated far more widely, with hallucination methods appearing in 37 of 66 studies, bias in 58 and safety in 61. The corpus-wide extraction supports a claim the individual studies cannot, that high nominal coverage conceals a much smaller body of actual measurement.

6.1 Hallucination and factual grounding

Two studies operationalise hallucination as a measurable quantity, using incompatible definitions.

Linardon et al.[35] converts hallucination into a verifiable outcome, citation fabrication. Six literature reviews of roughly 2,000 words were generated in a 3 × 2 factorial design crossing disorder familiarity with prompt specificity. All 176 citations were manually verified across Google Scholar, Scopus, PubMed and WorldCat, then classified as fabricated, real-with-errors or accurate, with error subtypes recorded down to author, year, title, journal, volume, issue, page and DOI. Two-tailed chi-square tests of independence were used with pairwise comparisons. Fabrication affected 19.9% of citations overall but concentrated in the less-visible disorders (28% and 29% against 6% for major depressive disorder). Hallucination is thus conditional on informational terrain rather than a fixed model property, a pattern a single-condition design could not have detected. Scope is narrow: one model, one output per condition, and no test of mitigation.

Kim et al.[28] define a construct the factual literature does not capture. Affective hallucination denotes emotionally immersive responses evoking false social presence, a model simulating emotional capacity it does not possess. Prompts from five mental-health subreddits produced

AHaBench (500 prompts) and AHaPairs (5,000 preference pairs), assessed on a 7-point rubric across Emotional Enmeshment, Illusion of Presence and Fostering Overdependence. The automated judge is validated against clinical ground truth, with two licensed psychiatrists rating outputs and human-GPT-4o agreement at $r = 0.85$ against inter-human agreement of $r = 0.95$. Mitigation is examined alongside diagnosis, comparing DPO alignment against SFT, SFT+DPO and few-shot baselines across model sizes from 7B to 72B, with MMLU, GSM8K and ARC used to verify that alignment does not degrade general reasoning. Few other studies here test for capability regression following a safety intervention. Prompts derive from Reddit and carry a demographic skew, evaluation is single-turn, and author annotation bias cannot be excluded.

6.2 Bias and stigma auditing

Yeo et al.[65] built a closed-loop agent-to-agent design of unusual care. A GPT-4 conversational agent interacted with GPT-3.5 Digital Standardised Patients across 97 demographic combinations of age, sex, race and income, each tested four to five times alongside controls, producing 449 transcripts and 4,502 agent responses. Information was made asymmetric: simulated patients remained agnostic to their own sociodemographic traits while the agent was informed of them through a separate channel, so any linguistic difference can be attributed to the agent. Measurement used LIWC-22 across 13 primary and 30 exploratory markers, avoiding human raters to remove researcher subjectivity. Analysis spans normality testing, subgroup comparison, multivariate regression, Joinpoint regression for piecewise trend analysis of tone across ten sequential dialogue bins, and group-based trajectory modelling. Analysing tone trajectory within conversations, rather than aggregate difference, captures bias emerging over an interaction. Patients are simulated, a single English-language model is examined, and LIWC may not detect subtler implicit bias.

Moore et al.[40] examined stigma and clinical appropriateness together. A mapping review of ten clinical guideline documents yielded 17 features of good therapy, used both to construct a steel-man system prompt, giving models their best chance rather than a strawman, and as the evaluation rubric. The first experiment applied modified National Stigma Studies vignettes, 72 vignettes across four conditions with 14 questions, measuring willingness to interact socially with the described character. The second tested responses to ten acute stimuli including suicidal ideation and delusions, with and without transcript context. Analysis used bootstrapped 95% confidence intervals with Bonferroni-corrected t-tests and z-tests of proportion, and inter-rater reliability reached Fleiss' $\kappa = 0.96$. Sixteen human therapists supplied baseline responses. Failing to challenge delusions, or supplying suicide-enabling information, was classified as unsafe.

6.3 Crisis-safety and escalation testing

McBain et al.[39] conducted the largest safety audit in the corpus by response volume. Thirteen clinical experts stratified 30 suicide-related questions into five risk levels using mean expert scores with explicit banding. Each of three commercial chatbots was queried 100 times per question, yielding 9,000 responses, each independently coded by two researchers as direct or indirect with third-coder adjudication, and indirect responses further classified by referral type. Mixed-effects logistic regression modelled the likelihood of a direct response against risk category, with query iteration as random effect and model as fixed effect, the correct specification for this clustered design. Stratifying refusal behaviour across five clinician-calibrated levels rather than a binary safe/unsafe split surfaces inconsistency at intermediate risk that a binary framing would not detect.

Heston[19] is the smallest study in this section. All 25 conversational agents identified on a public repository were tested twice, once with general distress prompts and once with PHQ-9-anchored prompts, recording the severity point at which each agent referred to a human counsellor and the point at which it shut down. Anchoring escalation to PHQ-9 renders slow escalation quantifiable in clinically interpretable units. Simulation continued past shutdown to test restartability, and 22 of 25 safety-triggered agents resumed normal conversation once distress severity dropped, which indicates that the shutdown state was not durable, a behaviour a single-pass test would not detect. Shutdown points seem to be governed by general-purpose guardrails rather than agent-specific prompting. Limitations include a single evaluator, fixed prompts, descriptive statistics only, and agents of an earlier model generation.

Campbell et al.[8] conducted the corpus's only longitudinal safety study, collecting responses in Spring 2023 (5 prompts, 3 chatbots) and again in Summer 2024 (7 prompts, 7 chatbots). Two trained school counsellors with sixteen years' combined adolescent crisis experience coded responses for empathy level, presence of crisis resources, tone, lethality handling and urgency, with disagreements resolved by consensus, and LIWC-22 supplied authenticity and emotional tone scores. Two design choices reflect the target user population: only free accounts were used, matching adolescent access conditions, and prompts were framed as seeking help for a friend, a proxy framing that also bypasses direct safety triggers. Verification extends to whether the correct hotline number was given. The acknowledged gap is that direct first-person suicidality was never tested.

6.4 Adversarial and automated red-teaming frameworks

Three studies build reusable infrastructure for eliciting failure rather than measuring it on a single occasion.

Au Yeung et al.[66] introduced psychosis-bench, running 128 experiments across 1,536 conversational turns. Scenarios pair a delusional theme with a related harm type and appear in explicit and implicit variants, with the first three turns held identical between pairs so that divergence is attributable to cue framing alone. Three orthogonal metrics are scored by an LLM judge: Delusion Confirmation Score, Harm Enablement Score and a binary Safety Intervention Score. Paired t-tests support the implicit-versus-explicit contrast and Spearman's rank-order correlation relates harm enablement to delusion confirmation. Scenarios were clinician-written and validated, though scoring was performed entirely by the LLM judge with no reported human validation, which limits interpretation given that Kim et al.[28] show such validation is achievable. Coverage extends to 16 scenarios and 12-turn conversations.

Steenstra et al.[56] developed a multi-session simulation framework. Fifteen clinically validated personas, five alcohol use disorder phenotypes crossed with three stages of change, were instantiated as patient agents with an embedded dynamic cognitive-affective model, managed by an orchestrator persisting state across checkpoints. A full factorial design crossed six therapist conditions with 30 stratified patient pairings across four longitudinal weekly sessions, giving 180 dyads and 369 completed sessions. A four-stage repeated-measures cycle tracked pre-session outcomes, in-session crises, post-session alliance and fidelity, and between-session narratives. Analysis used linear mixed-effects models for longitudinal metrics, generalised linear models for count safety outcomes, and 1,000-iteration bootstrapping for saturation analysis, reporting that 9.68 pairings on average suffice to reach 95% metric saturation, which provides a design parameter for replication. Human validation covered clinical realism and dashboard usability with nine assessors each. A deliberately harmful AI condition was included alongside commercial systems, providing a lower anchor that most evaluations omit.

Cai et al.[7] developed PsyCrisis for reference-free safety evaluation in Chinese. A curated dataset of 608 real crisis utterances covering suicide, non-suicidal self-injury and existential distress was drawn from online discourse, with risk categories defined using WHO mhGAP and LIVE LIFE guidelines. Eight hundred open-ended responses were scored across five binary dimensions using psychologist-verbalised chain-of-thought reasoning, so each judgement traces to an expert-defined criterion rather than a holistic score. A reference-free design avoids the impracticality of curating gold answers for every possible crisis utterance. Six clinical professionals annotated categories, scores and explanation preferences, with agreement reported as Cohen's κ

= 0.697, F1 = 0.802 and Fleiss' κ. Annotator compensation is disclosed, which no other study in the corpus reports. Scope is single-turn and Chinese-only, and leniency bias in the evaluator cannot be ruled out.

6.5 Corpus-wide appraisal: coverage without measurement

The aggregate picture across all 66 studies differs from that suggested by the ten studies above.

Table 6.1: Risk evaluation across all 66 studies

Dimension	Studies with content	Share	Of which: actual measurement
Safety	61	92%	11 test output safety or crisis handling
Bias	58	88%	5 apply counterfactual, linguistic, or stigma-probe methods
Hallucination	37	56%	11 score or verify it

Safety coverage is dominated by data governance rather than output testing. Of 61 studies with safety content, 26 describe research-ethics or data-governance measures such as IRB approval, consent, encryption, regulatory compliance and de-identification, and 24 discuss safety narratively. Eleven actually test whether model output is safe. The two answer different questions: encrypting a transcript protects participant privacy without indicating whether the system will supply harmful information to a person in crisis. Reporting that 92% of studies address safety, without this decomposition, would be misleading.

Bias is overwhelmingly acknowledged rather than measured. Of 58 studies with bias content, 13 note bias as a concern without evaluating it and 40 offer descriptive or incidental treatment. Five apply a recognisable bias method: counterfactual manipulation, linguistic instrumentation and stigma-vignette probing. Given that the corpora characterised in Section 2 derive largely from English-language Western platforms, and that Song et al. report cultural mismatch as a leading user complaint, five measurement studies across 66 is a thin evidential base for claims about fairness.

Hallucination has the lowest coverage but the healthiest measurement ratio. Of 37 studies, 11 measure or score it, 5 mitigate architecturally without measuring, and 21 discuss it only. The mitigation-without-measurement group is the one flagged in Section 3, where retrieval augmentation is invoked as a hallucination control and never tested.

The two hallucination constructs do not compose. Linardon et al. measure factual fabrication and Kim et al.[28] measure affective simulation. A system could score well on the first while failing on the second, and a heavily guardrailed system that never fabricates a citation may be the one that over-performs emotional presence. No framework currently integrates factuality, faithfulness and affective hallucination into a single assurance account.

Crisis findings run counter to the triage findings of Section 5, and the divergence is likely methodological. The triage studies found conservative over-escalation whereas the crisis studies here find under-escalation, non-durable shutdowns and inconsistent guardrails at intermediate risk. The explanation lies in stimulus design: triage studies present a complete case and ask for a rating, whereas crisis studies present an escalating user in distress and observe behaviour. Single-turn rating tasks do not surface the failures that multi-turn escalation produces, which supports the use of multi-turn designs.

Human validation of automated judges is inconsistent even within this section. Kim et al.[28] validate their judge against two licensed psychiatrists, Cai et al. against six clinical professionals, and Au Yeung et al. report no human validation of judge scoring. Since all three use LLM judges to produce safety verdicts, any leniency bias in the judge, which Cai et al. identify as a limitation, propagates into the safety conclusion.

Table 6.2: Design characteristics of the ten risk-primary studies

Study	Turn structure	Stimulus source	Judge	Human validation of judge	Inferential statistics
Linardon et al.[35]	Single output	Generated reviews	Human (authors)	Not applicable	Chi-square, pairwise
Kim et al.[28]	Single-turn	Reddit-sourced	GPT-4o	2 licensed psychiatrists	Pearson r, MAE, 3 seeds
Yeo et al.[65]	Multi-turn simulated	Agent-to-agent, blinded patients	LIWC-22 (automated)	Not applicable	Joinpoint regression, GBTM, multivariate
Moore et al.[40]	Single + context variants	Guideline-derived, stigma vignettes	Classifier	Psychiatrist + CS validation; Fleiss' $\kappa = 0.96$	Bootstrap CIs, Bonferroni
McBain et al.[39]	Single-turn $\times 100$	Expert-stratified queries	2 human coders + adjudicator	Not applicable	Mixed-effects logistic regression
Heston[19]	Escalating multi-turn	PHQ-9-anchored simulation	Single human (author)	Not applicable	Descriptive only
Campbell et al.[8]	Single-turn, 2 phases	Proxy-framed prompts	2 school counsellors	Consensus resolution	LIWC percentiles
Au Yeung et al.[66]	12-turn escalation	Clinician-written scenarios	LLM judge	None reported	Paired t-tests, Spearman
Steenstra et al.[56]	4 longitudinal sessions	15 validated personas	Automated + orchestrator	9 psychology professionals	LMM, GLM, bootstrap saturation
Cai et al.[7]	Single-turn	608 real crisis utterances	GPT-4o, expert CoT	6 clinical professionals	Correlations, κ, F1, majority vote

7 Cross-Cutting Methodological Quality

This section examines the corpus by quality rather than contribution type, reporting across all 66 studies using the extraction fields directly. The purpose is to convert the preceding account into an evidence-based statement of what this literature can and cannot support. One caveat applies throughout: the appraisal draws on what studies report and is not a validated risk-of-bias instrument, so under-reporting and under-performance cannot be distinguished. Where reproducibility depends on reporting, an unreported method is for practical purposes an unavailable one.

7.1 Statistical rigour and inference

Fewer than half the corpus supports its claims inferentially.

Table 7.1: Statistical treatment across 66 studies

Treatment	Studies	Share
Inferential statistics reported	28	42%
Descriptive statistics only	31	47%
No statistical analysis reported	7	11%

The 31 descriptive-only studies are not uniformly at fault, since the ten reviews and conceptual papers have no primary data to test and several qualitative studies report thematic analysis appropriately. The group also contains benchmark comparisons, model evaluations and system studies where performance differences are asserted across conditions without any test of whether those differences exceed chance. Where a study reports that one prompting strategy or model outperformed another without inferential support, the resulting ordering should be treated as provisional.

Table 7.1b: Distribution of statistical tests across the corpus

Test family	Studies
t-tests	10
Regression (linear, logistic, hierarchical)	6
Chi-square	5
ANOVA	4
Bootstrap	4
Mixed-effects / multilevel models	5
Kruskal-Wallis	3
Explicit confidence intervals	3
Multiple-comparison correction (Bonferroni)	2

The distribution suggests a narrow standard toolkit rather than method matched to design. Two studies report multiple-comparison correction, despite many conducting dozens of comparisons across models, prompts and metrics, and three report explicit confidence intervals on their headline estimates, so most reported differences carry unquantified uncertainty. Where studies repeatedly query the same model on the same items, a structure inherent to almost every benchmark and vignette design here, mixed-effects specifications are the appropriate response to the resulting clustering, and five studies use them.

Table 7.2: Methodological rigour by section

Section	n	Inferential statistics	Reliability coefficient	Code/data released
1 Reviews and conceptual	10	0 (0%)	2 (20%)	2 (20%)
2 Data foundations	12	4 (33%)	2 (17%)	4 (33%)
3 Model adaptation	14	3 (21%)	0 (0%)	5 (36%)
4 Benchmarks	6	5 (83%)	3 (50%)	1 (17%)
5 Human and clinical evaluation	14	6 (43%)	5 (36%)	2 (14%)
6 Risk and assurance	10	8 (80%)	2 (20%)	1 (10%)

This gradient matches the section-level observations reported above. Adaptation studies report the least inference and no reliability coefficients, while benchmark and risk studies report the most. The inverse relationship between engineering scope and evaluative depth noted in Section 3 is visible at corpus level rather than being an artefact of selective reading.

Table 7.3: Rigour by publication year

Year	n	Inferential statistics	Reliability coefficient
2023	4	0 (0%)	0 (0%)
2024	17	4 (24%)	1 (6%)
2025	36	16 (44%)	9 (25%)
2026	9	6 (67%)	4 (44%)

Inferential reporting rises from zero to 67% and reliability reporting from zero to 44% across four years. Much of the weakness documented above is concentrated in earlier work. This is consistent with the observation in Section 1 that prior reviews, whose search windows close by August 2024, describe a methodologically weaker literature than the one now current.

7.2 Reliability and inter-rater agreement

Fifty-nine of 66 studies involve human evaluation and fourteen report any reliability coefficient.

Table 7.4: Reliability statistics reported

Coefficient	Studies
Cohen's κ	7
Fleiss' κ	5
Krippendorff's α	2
ICC / intraclass	2
Cronbach's α, McDonald's ω, AVE	1
Test-retest correlation	1
Any coefficient (unique studies)	**14 / 66 (21%)**

The relevant figure is the intersection: 45 of the 59 studies using human evaluation report no agreement statistic. Human judgement serves as ground truth in most of this literature, with 33 of 59 studies using licensed clinical professionals, yet in three quarters of cases rater consistency cannot be assessed. Where a study reports that clinicians rated model responses favourably without agreement data, an unquantified assumption of rater consistency is doing the work. Reported values range from Fleiss' $\kappa = 0.96$ down to 0.613 inter-annotator and Cohen's $\kappa = 0.566$ for LLM-human agreement in harm-log coding[41], with $\kappa = 0.697$ elsewhere. Coefficients in the 0.55 to 0.70 band are moderate, and the studies reporting them state as much; the 45 studies reporting no figure remain unassessed.

Two procedures reported in Sections 4 and 5 remain uncommon elsewhere in the corpus. Separating consistency ICC(C,1) from absolute agreement ICC(A,1) distinguishes a rater who ranks correctly from one who is correctly calibrated, and applies to human panels as much as to LLM judges, though no human-rater study in this corpus reports it. The psychometric apparatus applied to an LLM-derived measure, covering α, ω, AVE, CFA fit indices and bootstrap optimism correction, indicates that classical test theory is applicable, and appears once.

7.3 Reproducibility and artefact availability

Table 7.5: Reproducibility markers across 66 studies

Marker	Studies	Share
Code, data, or models released	15	23%
Hyperparameters specified	14	21%
Hardware specified	13	20%
Decoding parameters specified	13	20%
Reporting standard followed (PRISMA, STROBE, CONSORT, MMAT, RoB 2)	12	18%
Protocol preregistered	2	3%

Roughly one study in five documents the parameters required to reproduce it. Decoding parameters are the principal omission: temperature, top-p and sampling settings change model output, and 53 of 66 studies do not report them. Since repetition is inconsistently applied and confidence intervals are rare, a share of reported performance differences cannot be distinguished from sampling variance. Studies that fix temperature at 0 for determinism, or report full decoding configurations, indicate that the reporting cost is low. Preregistration at 3% is low relative to adjacent fields, given that 28 studies conduct hypothesis tests. The finding reported in Section 1 that models and prompts are underspecified to the point where replication is frequently impossible is independently corroborated by these figures.

7.4 Systematic design gaps

Four gaps recur across the corpus and are not attributable to individual studies.

Single-turn evaluation dominates a multi-turn problem. Of 56 empirical studies, 16 employ multi-turn, longitudinal or repeated-measures designs, nine explicitly state single-turn scope, and most of the remainder are single-turn by construction. Triage studies presenting a complete case report conservative over-escalation, while crisis studies presenting an escalating user report under-escalation, non-durable shutdowns and inconsistent guardrails. The restartability result, that 22 of 25 agents resume after a safety trigger once distress subsides, would not be detected in a single-pass design. Therapeutic relationships are longitudinal; the evaluation apparatus largely is not.

The prospective evidence base is one underpowered trial, two weeks, $n = 160$, explicitly not powered for between-group inference, conducted by the product developer and excluding participants with recent suicidality. No study in this corpus reports clinical outcomes from an adequately powered trial, so every efficacy claim rests on proxy measures, whether benchmark accuracy, expert appropriateness ratings or satisfaction scores, none of which has been validated against patient outcomes.

Linguistic and cultural coverage is narrow and unevenly examined. Chinese-language work is well represented with 10 studies, with smaller contributions in Italian, German, Spanish and Indonesian. Four studies explicitly note English-only scope as a limitation, the dominant corpora are English-language Western platforms, and only five studies apply any recognisable bias method. The geographically diverse interview sample reporting cultural mismatch as a leading user complaint identifies a problem the rest of the corpus is not equipped to investigate.

Stochasticity is unaddressed in four studies out of five. Twelve of 66 address repetition, repeated trials or test-retest reliability, and the remainder report single-pass results from systems that are non-deterministic by default. Repetition yields more than error bars, since response consistency across trials predicts accuracy while self-reported confidence does not, providing a reliability proxy, yet the practice remains rare.

7.5 The gap addressed by this review

Six statements follow from the appraisal above and together define what this literature cannot currently support. No pooled effect estimate exists for any LLM mental health application, since all seven prior reviews synthesise descriptively and the heterogeneity of metrics, tasks and ground-truth definitions explains why. Human evaluation is the field's primary ground truth and is largely unvalidated, with 45 of 59 studies reporting no agreement statistic. Safety coverage is nominally near-universal and substantively narrow, with 61 studies addressing safety and 11 testing whether model output is safe. Bias is acknowledged far more often than measured, with 58 studies addressing bias and five applying a recognisable method. Reproducibility is the weakest dimension, with one study in five reporting decoding parameters, hyperparameters or hardware, and one in thirty preregistering. Clinical efficacy is unestablished, with one underpowered exploratory trial constituting the entire prospective evidence base.

The present review addresses the first four of these by extracting methodology rather than results uniformly across 66 studies with four dedicated risk fields, permitting the corpus-wide claims made in Sections 6 and 7 that no prior synthesis could support. It cannot address the last two, which require primary research rather than review.

What a stronger design would require follows directly: multi-turn or longitudinal interaction rather than single-turn stimulus; repeated trials with reported decoding parameters and confidence intervals; human evaluation with reported agreement, ideally separating consistency from absolute calibration; safety evaluation that tests output behaviour rather than documenting data governance; bias evaluation using counterfactual manipulation with clinically interpretable outcome instruments; and powered trials measuring patient outcomes rather than proxies. Each of these components already appears somewhere in this corpus, in longitudinal simulation, repetition protocols, ICC decomposition, escalation anchoring, counterfactual psychometrics and trial architecture, though not in combination.

The trajectory in Table 7.3 suggests such a combination may be closer than the aggregate figures imply. Reporting standards rise year on year, and the 2026 cohort reports inferential statistics at 67% and reliability coefficients at 44%, roughly triple and seven times the 2024 rates respectively. The methodological weaknesses catalogued here may therefore reflect the early stage of the literature rather than fixed properties of it.

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